

ELEN Design Project

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Abstract

This project aimed to design, construct and test a direct current (DC) motor capable of lifting a R5 coin to a vertical height of one metre. The motor converted electrical energy into mechanical energy through the interaction of electromagnetic induction. A sensitivity analysis was performed by varying the current and the gap size between the magnets to determine their effects on torque and motor speed. The results show that the motor operated most effectively at a current of 2.9A and a magnet gap of 33mm. This project illustrates how electromagnetism can be implemented to create a functional motor, highlighting the trade-offs between current consumption, coil count and motor efficiency. Ultimately, the project successfully met the aim of constructing a motor capable of lifting the R5 coin to a height of one metre.

1. Introduction

This project focused on designing, constructing and testing a DC motor that can lift a minimum load of an R5 coin through a vertical height of one metre. A DC motor converts electrical energy into mechanical energy through the interaction of electromagnetic induction, which enables it to perform work such as lifting a coin. The project explored various factors that influence the motor's performance, such as voltage, current, and magnet configurations. A sensitivity analysis was performed to assess the motor's performance by varying a single parameter at a time to determine the optimal configuration.

2. Aims and Objectives

Aim: To design, build and test a functional DC motor that is capable of lifting a R5 coin through a vertical height of one metre.

To apply the principles of electromagnetism and to investigate how various design parameters impact the performance of a DC motor.

Objectives: Perform a sensitivity analysis by changing one variable at a time (current and magnet gap size) and analyse how each variable affects the motor's ability to generate torque and lift a R5 coin.

Optimise and justify the final design based on the sensitivity analysis results and the efficiency of the DC motor.

3. Background Information

What is a DC Motor?

A DC Motor is an electrical device that converts electrical energy (direct current) into mechanical energy (Sytez, 2024). DC motors rely on electromagnetic induction, where a conductor carrying current (typically a coil of wire) in a magnetic field experiences a force that causes it to rotate (Sytez, 2024). It normally consists of two main components: a stator and a rotor. A stator provides a stationary magnetic field and a rotor, a coil of wire that rotates (Evans, 2020). A magnetic field is generated when a direct current flows through the coil (Evans, 2020). This magnetic field interacts with the stator, creating a force that causes the rotor to spin (Evans, 2020).

Key variables that affect the performance of DC Motors

The performance of a DC motor depends on key variables, such as the voltage, current, load, magnetic field strength and resistance. The specific impact of variables is:

- Applied voltages directly affect the speed of the motor (Islam, 2024). The speed of the motor increases with a higher voltage and decreases with a lower voltage.
- The current that flows through the motor directly influences the torque—increasing the current results in a greater torque, allowing the motor to handle larger loads (Csanyi, 2012).
- The speed and efficiency of a DC motor is impacted by a mechanical load attached to it. As the load increases, the motor requires a higher current to maintain its speed (Muller, 2016).
- The strength of the magnetic field (from permanent magnets or electromagnets) in the motor affects the force exerted on the rotor and, thus the torque and speed of the motor (Evans, 2020).
- The internal resistance of the motor's windings affects how much current flows for a certain voltage, affecting both the speed and torque of the motor (Evans, 2020). Higher resistance limits the current, reducing the performance of the motor.

4. Method and Apparatus

Basic Apparatus: Wooden Base, Copper Foil, Enamel Coated Copper Wire, Magnets, Wooden Dowel, String, Power Supply, R5 coin, Electrical Tape

Method:

Enamel Coated Copper wire was coiled and secured (with electrical tape) into a loop (about 3-4 cm in diameter) around the dowel, leaving 2-3 cm of straight wire on both sides of the loop. The straight ends were positioned directly opposite each other and soldered to copper foil in order to act as the split-ring commutator. Two loops were placed on the rotor and secured as shown in figures 1 and 2. The rotor was placed on the base, allowing for the dowel to rotate freely. Two magnets were secured on the base, positioned on either side of the rotor while ensuring the magnetic poles faced each other. Two thicker pieces of copper wire (brushes) were bent and secured into a shape that allowed them to press lightly against either side of the rotor on the copper foil, ensuring that the wire ends of the coil could make contact with them. A piece of string one meter in length was attached to the end of the dowel, and a R5 coin was attached to the other end. Each brush was connected to the positive and negative terminals of the power supply respectively.

A sensitivity analysis was conducted to determine optimum motor requirements. First, the current was varied within a circuit containing the motor.

$$V = IR \quad (1)$$

Using (1), the voltage was kept constant, and the current was decreased by increasing the resistance of the circuit by adding resistors in series, up until a point where the motor did not operate. The torque (velocity at which the coin was lifted) resulting from each current value was documented. Next, the gap size between the magnets was analysed by moving them closer together in consecutive intervals until they could not move any closer. The torque resulting from each gap distance was documented.

5. Results and Analysis

Figure 3 of the appendix shows how varying the current has an impact on the torque at which the motor can lift the load. These results were obtained by varying the resistance in series with the motor from no resistance up to 2Ω . It can be observed that these results are not linear, and level off as current increases above 2.9A. This indicated that the increase in current beyond a point did not significantly increase torque. This could be due to the additional current being converted into heat or that the rotor experienced higher friction as the motor spun faster. This indicated that there was a maximum efficiency of the motor, which was approximately at 2.9A.

Table 1 of the appendix shows how the decrease in magnet gap size increased motor speed. This was due to an increase in proximity between the magnets and the coils. As the magnets and coils were brought closer together, the magnetic field experienced by the coils was increased, thus resulting in an increased torque. It was then concluded that the minimum magnet gap size of 33mm was thus the optimal distance. This was fixed by the proximity of the magnets to the coils.

Through trial and error, it was found that the design could be optimised by reducing the weight of the rotor and maximising the number of coils. By doubling the number of coils, the motor was found to be able to spin at a voltage less than half that of the voltage required by the unoptimized motor. Figures 4-6 illustrate the first, second and third (optimum) motor iterations respectively. Longer magnets were procured, and it was found that the motor performance improved when the magnets were positioned horizontally, as a large proportion of the coils were within the magnetic field at a time.

The motor was operated using the power supplies supplied. However, it was observed that the current varied greatly between different voltages, thus rendering the obtained results nonsensical. As a result, it was decided that using a 12V battery in a circuit containing resistors that act as voltage shedders was the most reliable testing method.

6. Trade-offs and recommendations

A trade-off observed was the interaction between current and torque. An increase in current, which comes at a cost (electricity cost), resulted in an improved torque of the motor. A consideration for this trade-off would be to determine the required lift speed for the motor and to optimise the required current for this torque.

An additional trade-off was the number of coils vs the performance of the motor. It was observed that increasing the number of coils improved performance (inoperable motor vs a more efficient motor). More copper wire is required to increase the number of coils, which increases cost. A possible consideration is to optimise the motor design to a required lift speed and determine the minimal number of coils to achieve this requirement.

Based on the work achieved in this report, it would be recommended to optimise the number of coils and test a wider range of current values to truly optimise the current consumption of the motor, to reach a certain motor speed.

7. Conclusion

A working motor was produced that could lift a load of a R5 coin through a height of a metre. A sensitivity analysis was conducted, where current and magnet gap size was varied. It was found that an optimum was achieved by operating the motor at 2.9A with a magnet gap size of 33mm (the minimum gap size practically possible).

Appendix

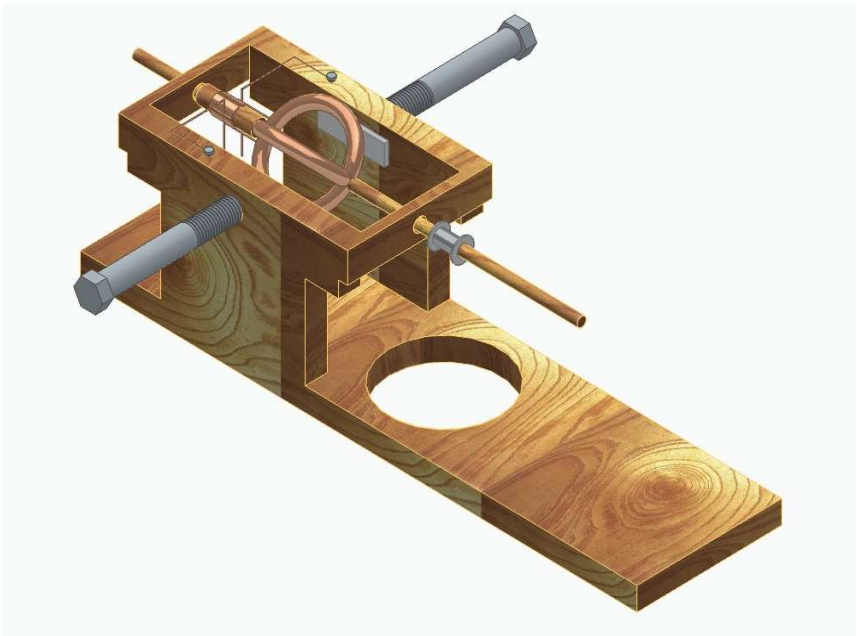


Figure 1: Isometric CAD construction of the optimised motor design

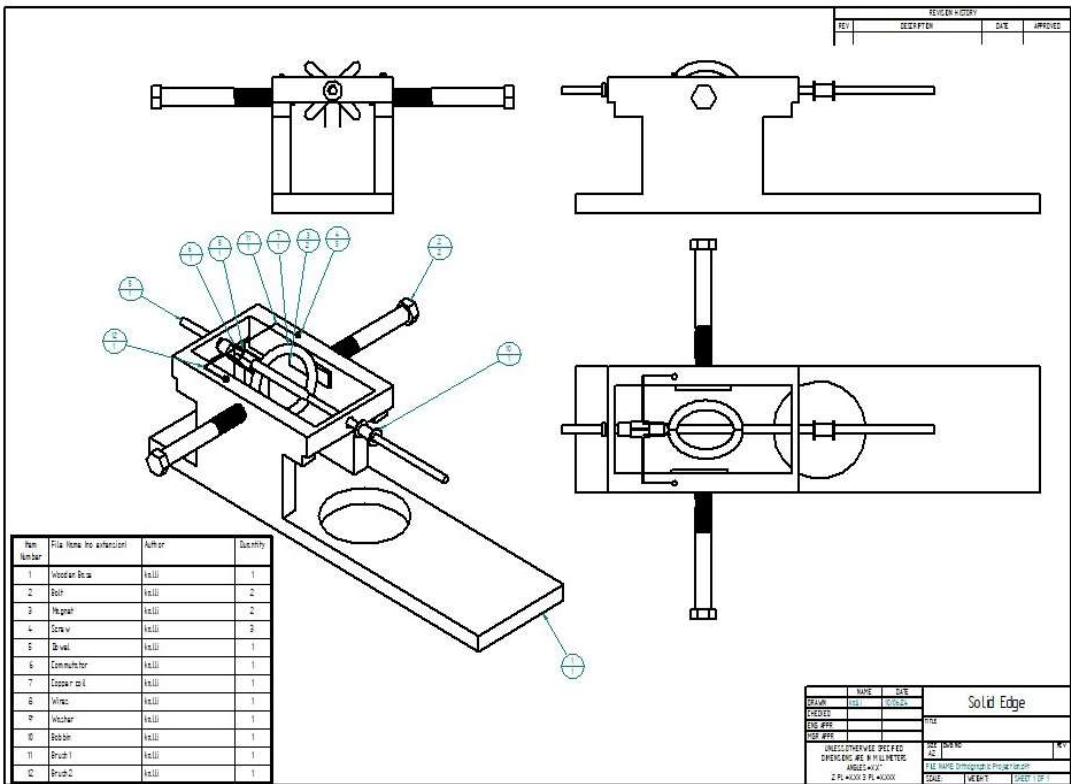


Figure 2: CAD construction of the optimised motor design, showing the different views.

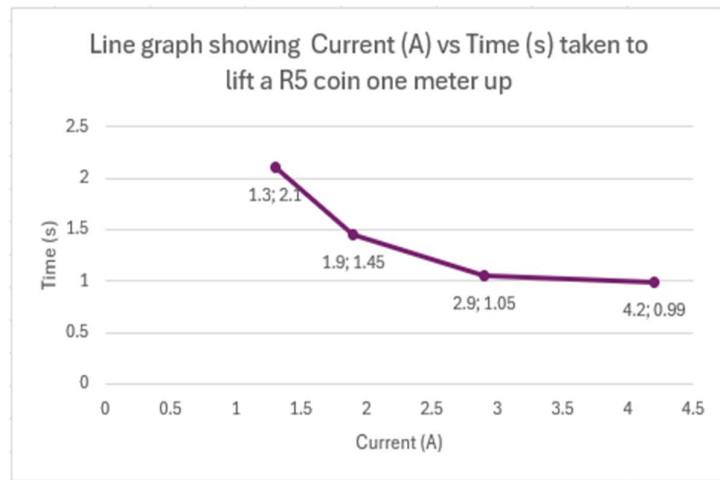


Figure 3: A line graph showing the current (A) vs the time (s) taken to lift the R5 coin one metre up, measured using a sensitivity analysis.



Figure 4: Iteration 1 rotor design



Figure 5: Iteration 2 rotor design

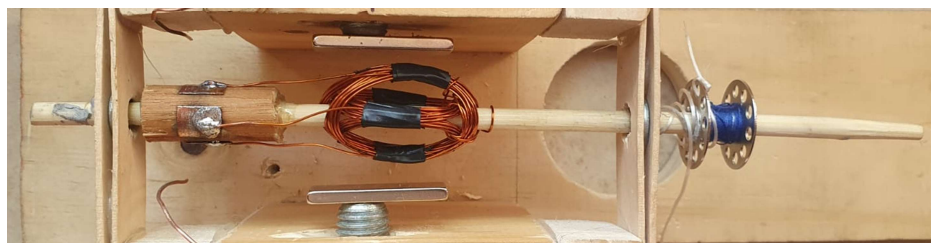


Figure 6: Iteration 3 rotor design (optimal design)

Table 1: A Table showing the results of the sensitivity analysis comparing the effect of different distances between the magnets on the time taken for the motor to lift an R5 coin one metre.

Distance between magnets (mm)	Time (s)
33	1.25
40	1.45
46	2.10

References

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Team contributions

Stator and Base build	Kallisha Naidoo (2803488)
Rotor build	Joanna Chronis (2817173)
Sensitivity Analysis	Kallisha Naidoo and (2803488), Joanna Chronis (2817173) (Done together)
Report (First Half, Abstract – Method)	Kallisha Naidoo (2803488)
Report (Second Half, Results – Conclusion)	Joanna Chronis (2817173)